Mplify Federation

Architecting the Global AI WAN

A strategic blueprint for transitioning isolated network capacity into a multi-trillion-dollar automated supply chain.

The Quality of Service Chasm: Why AI Remains Stranded

The Standard Internet	The AI Network Fabric
• Best-Effort Delivery	• Predictable Quality of Service (QoS)
• Outdated Routing Protocols	• Fully Programmable & Automated
• Rigid Capacity Limits	• Elastic & Massive Scale

$6.7 Trillion: Total data center investment by 2030 (McKinsey) cannot reach the physical edge without automated, scalable coordination.

The barrier to the global AI economy is network fragmentation.

Enterprise

Wholesale Carrier

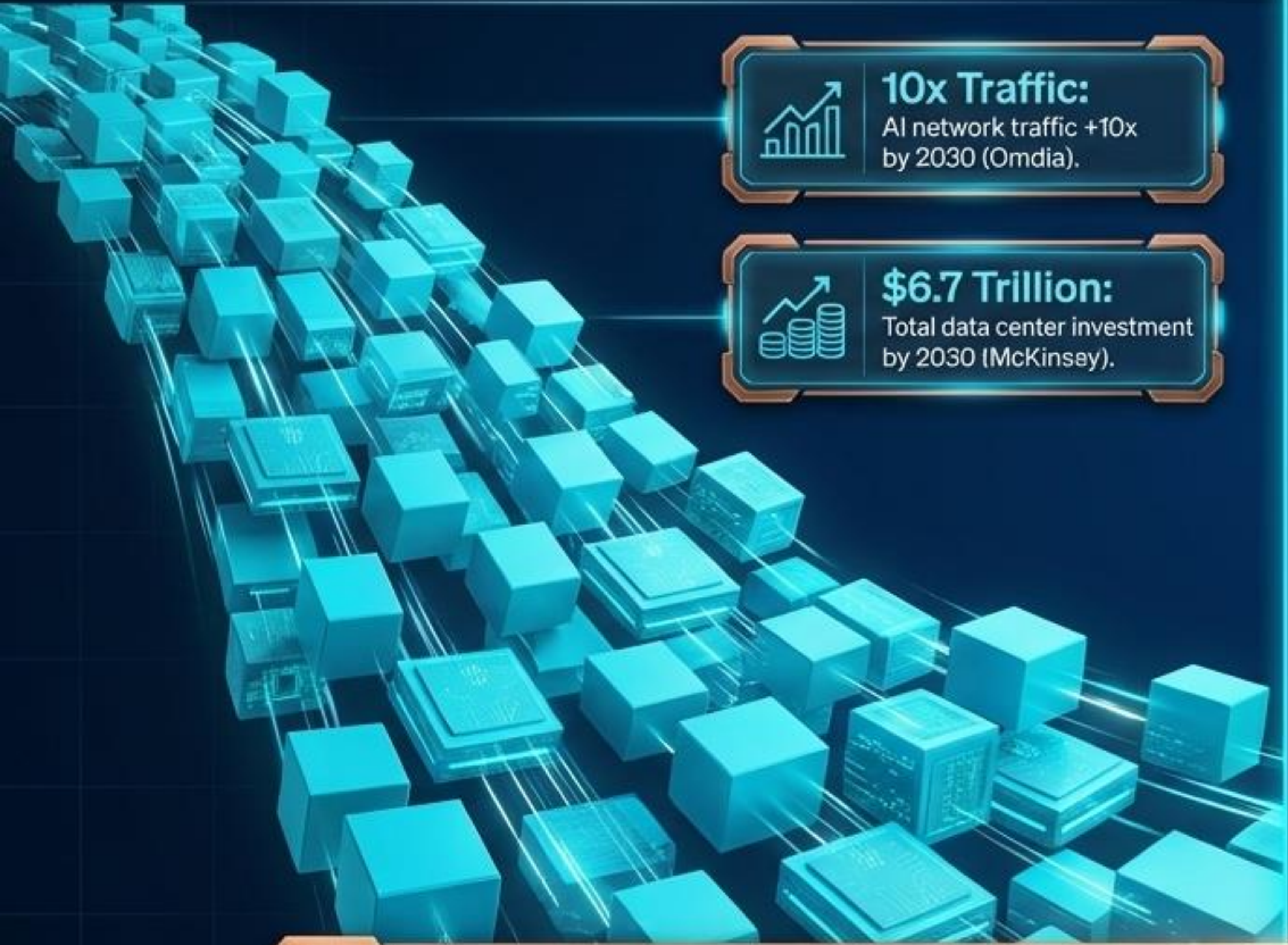
Cloud Provider

Service Provider

No single provider, carrier, or enterprise can build the massive, low-latency infrastructure required for AI alone. The capacity exists, but it remains dangerously siloed.

AI remains stranded without NaaS for AI.

The Demand: A Billion-fold Increase



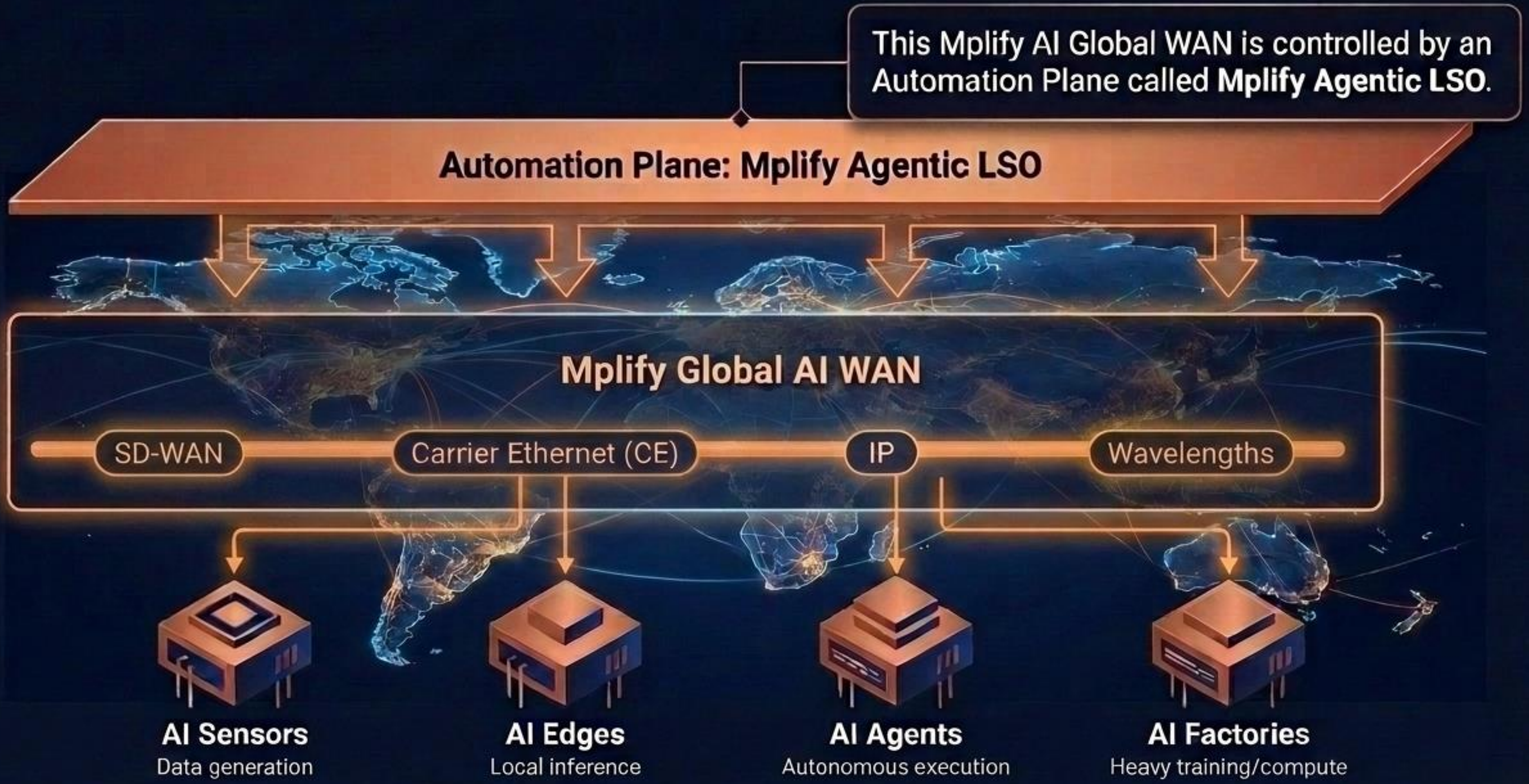
10x Traffic:
AI network traffic +10x by 2030 (Omdia).

$6.7 Trillion:
Total data center investment by 2030 (McKinsey).

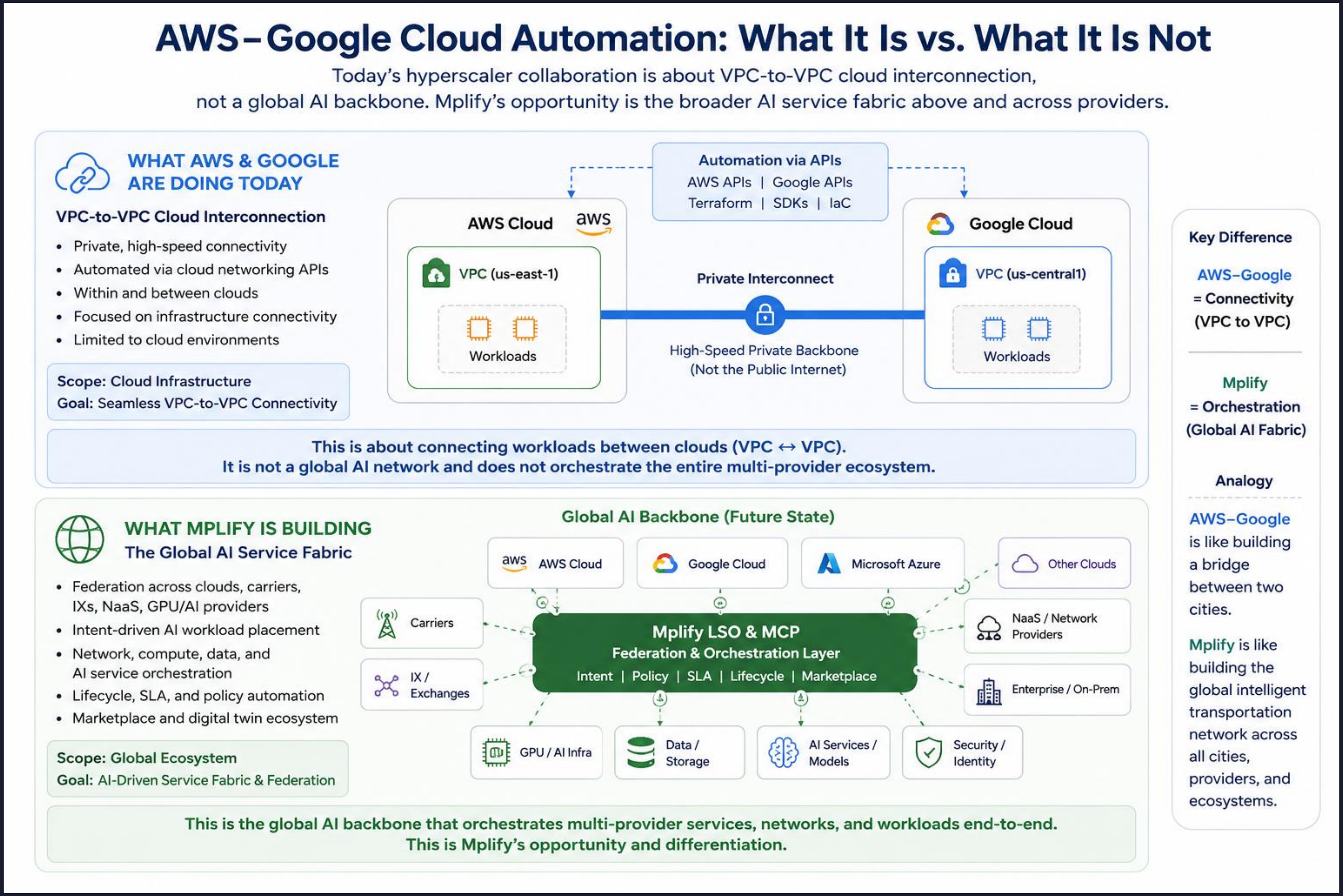
The Bottleneck: Custom Integration

Without automated, scalable coordination, the **$6.7 Trillion** data center investment cannot reach the physical edge. The network must evolve from a static utility into a dynamic supply chain.

Manifesting the Global AI WAN



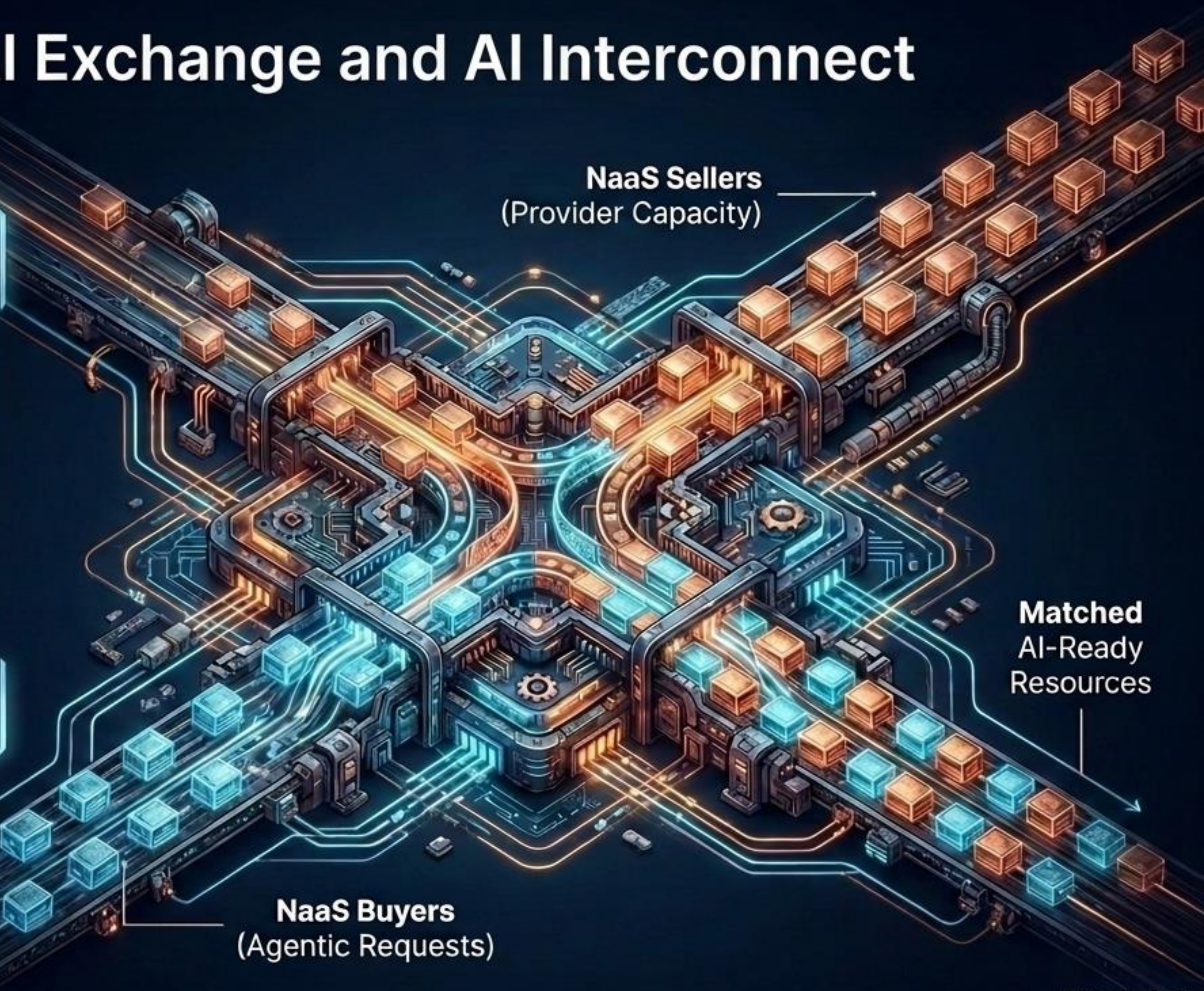
AWS–Google Cloud Automation vs. Mplify Global AI Fabric



The Global AI Exchange and AI Interconnect

Global supply (provider capacity) instantly meets demand (agentic requests).

A frictionless, programmable exchange where participants can dynamically buy, sell, and allocate AI-ready network resources without legacy bottlenecks.



The Mplify Federation Infrastructure Blueprint

Replaces bilateral custom integrations with a frictionless, programmable framework.

From Static Registry to Dynamic Digital Catalog

Replacing the legacy Mplify Member Registry with a live, machine-readable exchange where AI agents autonomously discover, collaborate, and provision ecosystem partners.

Tokenized Digital Twins

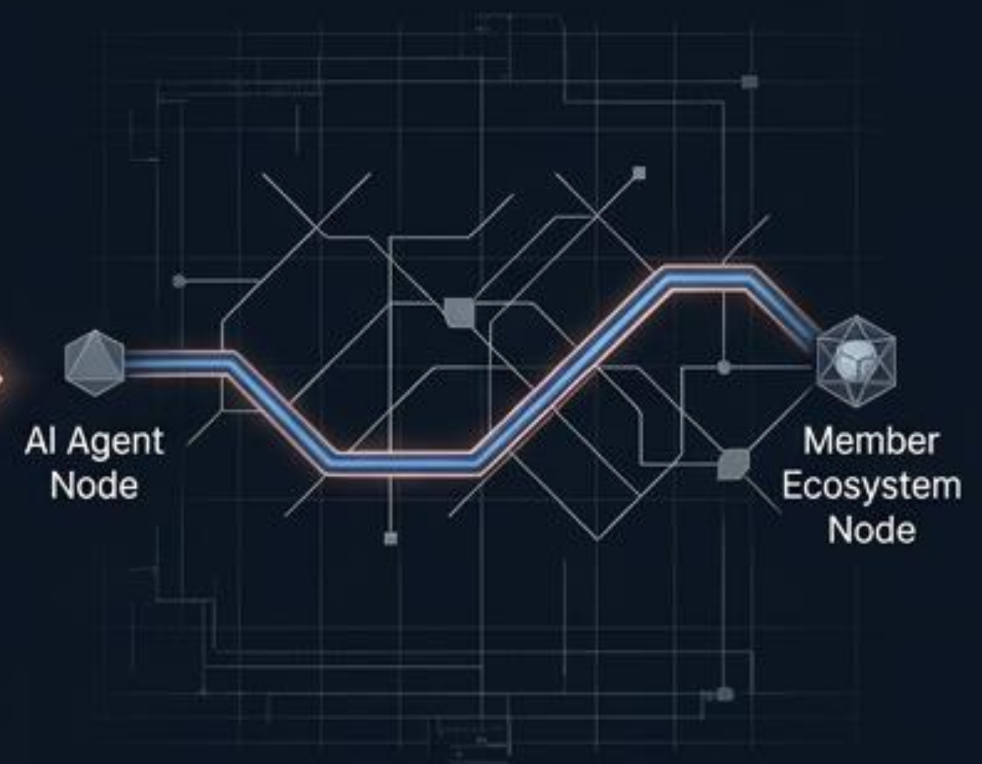
Members advertise their distinct capabilities as live, verifiable Digital Twins.

- Network-as-a-Service (NaaS) capacity
- LSO APIs (Cantata, Sonata)
- Automation protocols
- Zero-Trust Certifications

The Mplify Marketplace

A frictionless, programmable clearinghouse. It eliminates legacy bilateral integration bottlenecks, transforming into a live exchange where global provider capacity instantly meets agentic demand.

Collaborative AI Deals



Frictionless discovery translates into real-world business outcomes. Ecosystem partners easily find one another to instantly form on-demand supply chains, win joint bids, and provision collaborative AI infrastructure globally.

Federate and Monetize

The Final State

Full autonomy. AI Agents utilize MCP and LSO APIs to autonomously buy, sell, and route real-time inference workloads.

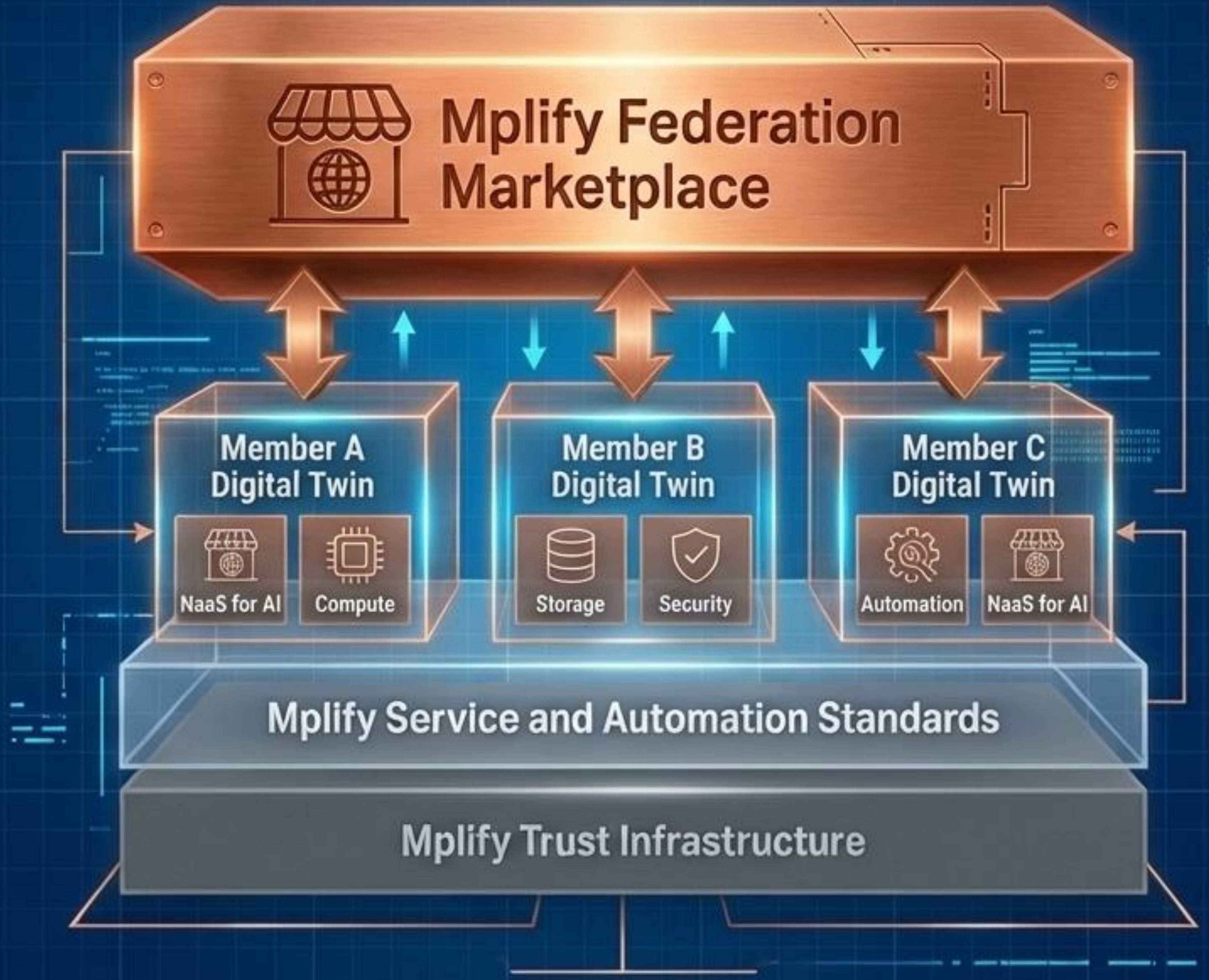
The Business Outcome

Telecom operators and enterprises stop being connectivity providers and become active participants in the AI economy, delivering AI-as-a-Service and capturing new revenue streams.

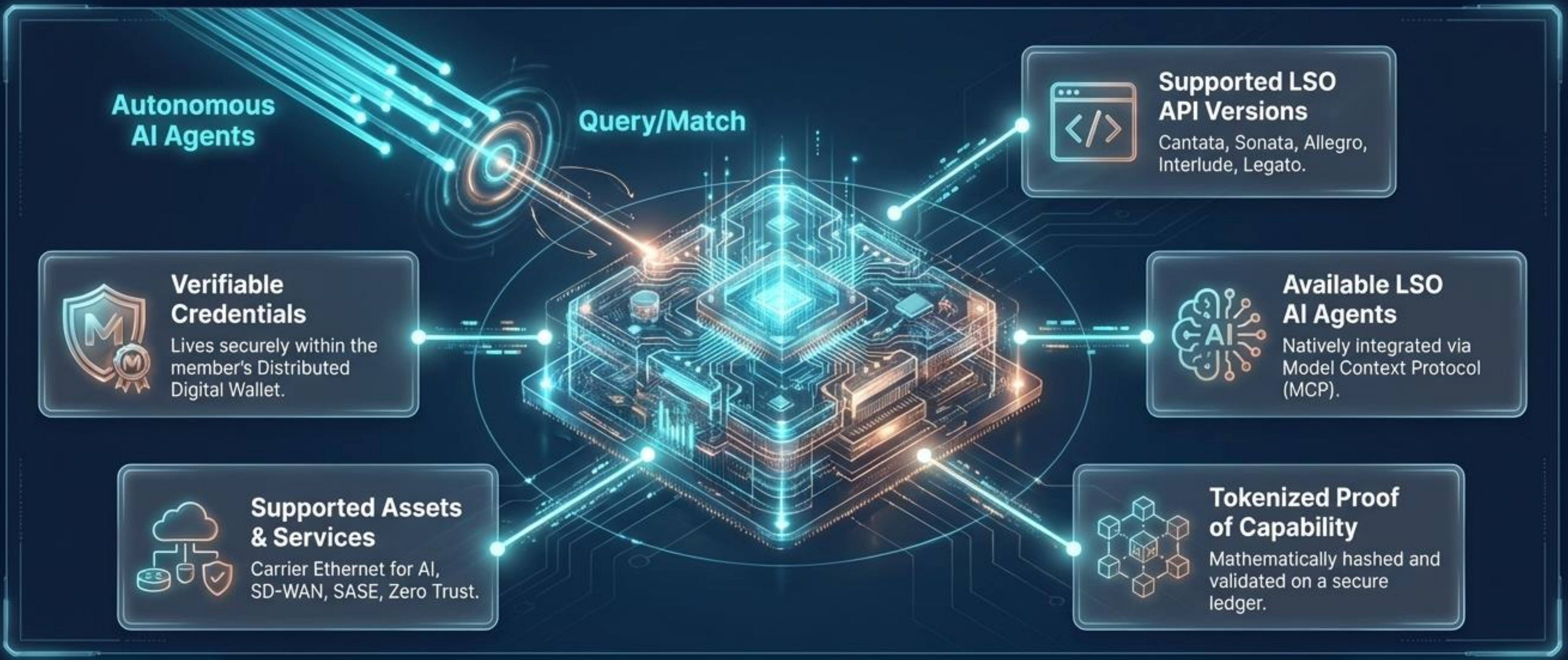
The Mplify Marketplace: Monetizing the Fabric

Each marketplace member has a unique 'Digital Twin' – a real-time digital representation of their capabilities.

These twins detail supported services, capacity, and real-time availability, enabling dynamic discovery and allocation.



The Mplify Digital Twin: The Live Proxy for Autonomous NaaS



Digital Twins are not static directories—they are live, machine-readable representations that agents autonomously query to source and provision physical infrastructure in milliseconds

The Prerequisite for Autonomy: Cryptographic Trust

The Paradigm Shift

To operate autonomously within the Federation, trust cannot rely on legacy human SLAs. It must be mathematically verifiable in real-time.

The Solution

The ecosystem relies on a decentralized, integrated trust framework. Provider capabilities, routing decisions, and hyper-distributed AI inferences are mathematically verified before execution.

The Mplify Federation: Strategic ROI for Members

Transforming isolated capacity into a unified, high-margin AI economy.

1. Escape the "Dumb Pipe" Trap

From Connectivity to AI Platforms

Own the Value Chain:
Pivot from selling generic bandwidth to delivering high-margin, outcome-based AI intelligence.

Edge Domination:
Transform existing telecom infrastrueture into geographically distributed AI performance platforms.

Reclaim the Developer:
Prevent OTTs and hyperscalers from capturing the AI edge by offering a superior, integrated developer ecosystem.

2. Monetize the AI Fabric

The Frictionless Exchange

Marketplace Realization:
Dynamically allocate and sell AI-ready network capacity on-demand via the Mplify Marketplace.

New Revenue Streams:
Expose real-time network intelligence through AFI monetization (e.g., predictive quality, fraud scoring).

Sovereign AI Clouds:
Capture highly-regulated enterprise and government sectors by guaranteeing localized data governance.

3. Elevate Operational Efficiency

Autonomous, Zero-Trust Networks

Predictable QoS:
Guarantee the strict, deterministic latency and high-throughput required for massive machine-to-machine AI uplinks.

AI-Driven Autonomy:
Slash OPEX utilising end-to-end network automation, predictive maintenance, and self-healing digital agents.

Embedded Security:
Enforce Zero-Trust cryptographic verification in real-time, eliminating vulnerabilities across distributed inference nodes.

4. Capture First-Mover Advantage

Instant Global Interoperability

Bypass the Build:
Avoid the prohibitive CapEx and multi-year delays of building proprietary AI infrastructure from scratch.

Eliminate Custom Integration:
Utilize LSO APIs and Model Context Protocols (MCP) te instantly piug into a global, standardized supply chain.

Cure Piiot Paralysis:
Deploy proven, pre-validated NVIDIA and Cisco ERA-compliant architectures to move instantly from concept to production.

The Enterprise Advantage: Sourcing AI from a Mplify Member who is part of the global Mplify federation

Title:
Frictionless, On-Demand Procurement

The Shift:
Manual API Mapping → Autonomous AI Agents

Subtext:
Dynamically source network and compute assets via native Model Context Protocol (MCP) intogration, eliminating the legacy burden of custom intagrations.

Title:
Accelerated Time-to-Value

The Shift:
Deployments → Deployment in Minutes/Days

Subtext:
Slash time-to-market by leveraging Mplify NaaS certification and standards, completely bypassing fragmented, from-scratch infrastructure builds.

Title:
Predictable, Deterministic Performance

The Shift:
Best-Effort Transport → Guaranteed QoS & Ultra-Low Latency

Achieve the stringent millisecond precision required for real time Physical AI and robotics, powered by RDMA over Converged Ethernet (RoCE).

Title:
Embedded Zero-Trust Security

The Shift:
Perimeter Defense → Mathematical, Localized Trust

Subtext:
Ensure absolute data sovereignty with policy-enforced, private pathways that isolate AI models natively and verify capabillties before execution.

Title:
NaaS for AI

The Shift:
Static Connectivity → Elastic AI Supply Chain

Subtext:
Transform isolated network capacity into a fully programmable, distributed compute substrate engineered specifically for the demands of the AI economy.

The Federation Infrastructure Fabric: Building Your Own Ecosystem

Join the Mplify federation to build the standard for the AI economy.

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